

GitHub: [sahajivvix-1.github.io](https://github.com/sahajivvix-1)

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Sahaj Saliya

Portfolio: sahajivvix-1.github.io/Portfolio2026

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Education

Pandit Deendayal Energy University (PDEU)

B.Tech in Information & Communication Technology (ICT), CGPA: 8.5/10.0

Gandhinagar, Gujarat

June 2023 – May 2027

Experience

Research & Innovation Intern

May 2026 - Jul 2026

HNNOIX Private Limited

- Worked on **6G research and development**, focusing on network architectures, AI agents, and communication systems.
- Developed reusable AI components including the **LLM Gateway**, **RAG Runtime**, **AI Memory Layer**, and **Knowledge Base** for agent execution.
- Implemented **Generative AI**, **LLM**, **RAG**, and **agent-based** solutions for decision-making, retrieval, and automation in **6G networks**.

Freelance Financial AI Engineering

Dec 2024 - Jun 2025

Quantitative Investment Analysis

- Developed **2 autonomous trading signal systems** using **Agentic AI** and **RAG** to analyze market news, extract sentiment, and generate structured trading signals.
- Reduced false-positive trade signals by **37%** by applying filtering rules and confidence scoring to remove low-confidence signals from the final output.
- Built a **CrewAI** research pipeline using **GPT-4**, **Bloomberg API**, **Financial Modeling Prep**, and **Alpha Vantage** for earnings analysis and market research.
- Developed a **Weaviate RAG pipeline** for semantic search over earnings transcripts, retrieving relevant financial information for analysis of company performance and financial statements.

Selected Projects

Enterprise Agentic RAG Orchestrator

Github

- Architected a production-grade **Multi-Agent RAG** system using **LangGraph** and **FastAPI** featuring a **Supervisor** → **Worker** → **Validator** pipeline to route queries across Corrective RAG, NL-to-SQL, and human-in-the-loop email tools.
- Engineered a self-correcting **Corrective RAG (CRAG)** loop with LLM document grading and query rewriting, utilizing **hybrid retrieval** (Qdrant dense + BM25 sparse), **CrossEncoder reranking**, and **parent-child chunk expansion** (200 → 800 tokens).
- Benchmarked system with **RAGAS** achieving **0.81 Faithfulness** and **1.00 Context Precision**; integrated sub-millisecond **Qdrant ANN semantic caching**, Redis session memory, prompt-injection guards, and **Fernet** encryption.

OpenEnv - Reinforcement Learning Data Cleaning Agent

Github

- Built a custom **Q-Learning (RL)** environment where a **Llama 3** agent autonomously selects cleaning actions for noisy tabular data using structured observations and reward feedback.
- Designed **reward shaping**, **anti-loop penalties**, and step constraints to reward missing-value reduction and deduplication while discouraging redundant or destructive operations.
- Implemented **Pydantic** schemas and an **LLM pipeline** with structured JSON outputs, then developed a **FastAPI** server with REST support and deployed it using **Docker** on **Hugging Face Spaces**.

Explainable Student Performance Prediction Platform

Github

- Built a **MLOps platform** for student-performance prediction with **Pandera** data validation, automated feature engineering, and benchmarking across **9 ML-DL models** including XGBoost, CatBoost, LightGBM, PyTorch, and FLAML.
- Implemented model explainability using **SHAP** and **LIME**, generating global and local feature attributions with Spearman rank correlation to measure explanation agreement.
- Integrated **MLflow**, **FastAPI**, and **Streamlit** for experiment tracking, deployment, and visualization, with **Docker Compose** and **GitHub Actions** for containerization and CI.

Technical Skills

Languages: Python, C++, SQL, MATLAB

AI/ML & GenAI: ML, DL, CV, NLP, Q-Learning, LLMs, RAG, Tool Calling & Function Calling, Multi-Agent Orchestration

AI Tools & Frameworks: LangChain, LangGraph, LLM APIs, AWS Bedrock, Pydantic

Vector Databases & Databases: ChromaDB, Qdrant, PostgreSQL, MySQL, MongoDB

MLOps & Deployment: MLflow, Docker, Kubernetes, Redis, Streamlit, Hugging Face Spaces

Software & Systems: Data Structures & Algorithms, Object-Oriented Programming, Git, Network Fundamentals & Troubleshooting & Analysis, Windows/Linux Administration, Routing & Switching

Certifications & Achievements

Agentic AI with LangChain and LangGraph : IBM

AWS Generative AI and AI Agents with Amazon Bedrock : Amazon Web Services (AWS)

Data Structure & Algorithm With Python : Udemy

Deep Learning : IIT Ropar (NPTEL); Score: 66%